OPEN IMMERSIONS DESCEND TO A FINITE LEVEL IN THE DESCENT SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Etale morphisms are open. Morphisms, Lemma morphisms-lemma-etale-open
- Descend isomorphism. Limits, Lemma lemma-descend-isomorphism (same chapter)

Source reference. Original reference 081D: sources/stacks-chow/tags.txt, source lines 10213–10213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07RP: sources/stacks-chow/tags.txt, source lines 9874–9874. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03WT: sources/stacks-chow/tags.txt, source lines 5124–5124. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 081E: sources/stacks-chow/tags.txt, source lines 10214–10214. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Notation and assumptions as in Situation 081D. If*

(1) *f is an open immersion, and*
(2) *f_0 is locally of finite presentation,*

then f_i is an open immersion for some $i \geq 0$.

Proof. By Lemma 07RP we can find an i such that f_i is étale. Then $V_i = f_i(X_i)$ is a quasi-compact open subscheme of Y_i (Morphisms, Lemma 03WT). let V and $V_{i'}$ for $i' \geq i$ be the inverse image of V_i in Y and $Y_{i'}$. Then $f : X \to V$ is an isomorphism (namely it is a surjective open immersion). Hence by Lemma 081E we see that $X_{i'} \to V_{i'}$ is an isomorphism for some $i' \geq i$ as desired. □

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